Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.
The journal volume, article number, publication date and DOI printed in this file are template placeholders, not a publication record.

TrueTap: Budgeting How Often a Smartwatch Buzzes You by Mistake

ANONYMOUS AUTHOR(S)

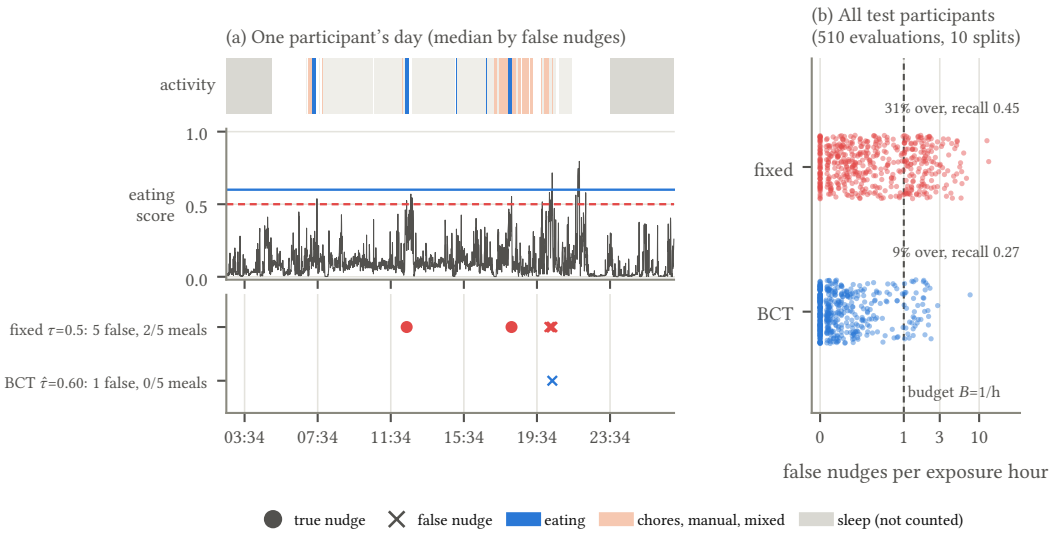


Fig. 1. A false-nudge budget on day-long wrist data (CAPTURE-24, eating, budget $B = 1$ false nudge per hour of exposure, that is, of labelled, non-target waking time). (a) A representative day: the test participant with the median number of false nudges under the conventional threshold in the first split. The detector was trained on 50 other people. With the conventional threshold ($\tau = 0.5$, dashed) the watch buzzes 5 times by mistake over the day and catches 2 of 5 meals. TrueTap's threshold, calibrated on 50 further people (solid), gives 1 false buzz and catches 0 of 5 meals. This participant was within budget even under the conventional threshold (0.42 false nudges per hour of exposure), so for them calibration cost meals without a benefit; the benefit appears across participants. Nudges minutes apart overlap at this scale; sleep and unlabelled time are not counted. (b) All 510 test-participant evaluations over 10 splits. The conventional threshold left 31% of new participants over budget (mean recall 0.45); the calibrated threshold left 9% (recall 0.27), for a target of at most 10%.

Smartwatches and research wearables increasingly buzz their wearers when they infer an activity: a workout, a drive, a meal, a Parkinsonian freeze. Their detectors are judged per window, but a wearer lives with false buzzes per hour. We define this burden per wearer, as false nudges per hour of non-target exposure after the full trigger policy, and present TrueTap, a layer between any activity detector and the vibration motor. TrueTap sets the trigger threshold by conformal risk control over calibration *users*, so that a new wearer from the same population, over a session like the calibration sessions, exceeds a chosen false-nudge budget with probability at most δ. It does not make a detector more accurate. It fixes the trade-off between false buzzes and missed events before deployment, with a stated risk.

On CAPTURE-24, a day of free-living wrist data from each of 151 adults, a conventional threshold left 7.6% of new participants over budget when pooled, which hid task–budget combinations with up to 31.4%. With calibration, the share was below 10% in every combination (pooled 3.7%, finite-sample bound 9.8%) and transferred across a mild shift between age and sex groups. The cost depended on the task: in-vehicle detection kept its recall (0.91 against 0.93 at one false nudge per hour), eating did not (0.27 against 0.45), and in 6 of 12

2026. ACM 2474-9567/2026/11-ART
https://doi.org/10.1145/nnnnnnn.nnnnnnn

combinations no threshold met both the budget and 80% recall. On three further datasets, window F1 tracked the 90th-percentile wearer's burden only moderately ($\rho = -0.55$). The guarantee is marginal over wearers, counts only nudges in labelled time, allows hourly bursts and breaks under a one-logit score shift; we map each of these limits.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; • **Computing methodologies** → *Machine learning*.

Additional Key Words and Phrases: smartwatch, activity recognition, just-in-time interventions, false alarms, haptic nudges, conformal risk control, evaluation methodology

1 Introduction

Smartwatches now act on what they infer about us. They start a hand-washing timer when they think we are scrubbing, ask whether we are working out, and call emergency services when they think we have fallen [5]. Research prototypes and habit-reversal bracelets vibrate when they detect face-touching, hair-pulling or other unwanted habits [41, 42, 66], and wearable systems cue people with Parkinson's disease when they detect a freeze of gait [6]. Each of these is an *activity-triggered haptic nudge*: a classifier's output, turned by a trigger policy into a buzz on someone's wrist.

The classifiers behind these nudges are judged in the classifier's terms: per-window accuracy, F1 or AUROC under leave-one-subject-out cross-validation [14, 53, 78]. The wearer notices something else: how often the watch buzzes when nothing is happening, how many real episodes it misses, and how late the buzz arrives. The two can be far apart. On the Daphnet freezing-of-gait data [6], a LightGBM detector reaches 91% window accuracy, barely above the 90% of a detector that never reports a freeze. With a standard trigger, 5 of the 10 patients receive no false cues and the others 8–30 per hour, while the patients who froze have, on average, 62% of their freezes go uncued. False buzzes have a cost: in a case report on a vibrating habit-reversal bracelet, a wearer stopped using it because everyday movements set off too many false alarms [42]. Yet no activity-triggered nudging system in HCI that we know of states, let alone controls, *how often* it will buzz a wearer by mistake. Thresholds and cooldowns are set by hand or by grid search.

We ask a simple question: *how often may a watch buzz a new wearer by mistake, and can that be promised before deployment?* We present **TrueTap**, a layer between any activity detector and the vibration motor (Figure 2). It counts false nudges per hour *for each wearer*, after smoothing, re-arming and cooldowns, and sets the trigger threshold on other users by conformal risk control with a user-level loss. A new wearer from the same population, over a session like the calibration sessions, then exceeds the budget with probability at most δ. We call this step Budget-Calibrated Triggering (BCT). Figure 1 shows what it changes for one participant's day and for every test participant.

We evaluate TrueTap offline on four public datasets: about a day of free-living wrist data from each of 151 adults in CAPTURE-24, scripted wrist sessions with built-in look-alikes in WISDM, free-living back and thigh data in HARTH, and ten patients with Parkinson's disease in Daphnet. Three findings stand out.

- **Window metrics are only a moderate guide to the burden.** Across 70 detector–task combinations, window F1 rank-correlated with the 90th-percentile wearer's false nudges per hour at $\rho = -0.55$. On scripted wrist tasks, the trigger policy moved the burden as much as the detector did.
- **Calibration keeps new wearers within budget where conventional thresholds fail, and pooled reporting hides the failures.** Pooled over CAPTURE-24's four tasks and three budgets, the conventional threshold left 7.6% of new participants over budget, but 31.4% for

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

eating and 25.1% for in-vehicle detection at one false nudge per hour. With BCT, the point estimate was below 10% in every combination (pooled 3.7%, bound 9.8%).

- **A budget picks an operating point; it does not improve the trade-off.** In-vehicle detection kept its recall (0.91 against 0.93); eating did not (0.27 against 0.45). BCT's recall was within 0.04 of the best global threshold of 0.5 or above, chosen in hindsight on the test participants, and for eating and sports no threshold met both a 10% risk and 80% recall.

Contributions.

(1) **A wearer-centred formulation of false interruptions:** false nudges per hour of non-target exposure, per wearer and after the full trigger policy, with the two properties that make it hard to certify: it is not monotone in the threshold, and it is only as fine as the exposure it is measured over. With it we re-evaluate 14 detectors on seven tasks.
(2) **Budget-calibrated triggering:** user-level conformal risk control over this post-trigger rate, with its assumptions, finite-sample guarantee and scope stated precisely, including what it does not cover (Table 1).
(3) **An empirical map of when such budgets help, cost or fail**, on day-long free-living wrist data, scripted look-alikes, non-wrist free-living data and a small clinical population, with group shift, hourly bursts, uncounted nudges, score drift, repeat violators and small calibration sets as stress tests (Table 4).

A Python library and scripts that regenerate every number, table and figure are provided as supplementary material. All results are offline; no person has yet worn TrueTap.

2 Related Work

2.1 Activity-Triggered Interventions on the Wrist

Smartwatch activity recognition has matured from fine-grained hand activity [39] to privacy-aware multimodal sensing [10, 53], robust gestures on commodity hardware [37] and on-watch recognition with sub-second onset latency [78]. Behaviour-specific detectors exist for nearly every moment at which a watch might nudge its wearer: eating [71], smoking [58, 67], hand-washing [5, 43, 54, 74], face-touching [17, 25, 66], body-focused repetitive behaviours [1, 61, 62, 68], freezing of gait with cueing [6, 47, 48, 69] and falls [12, 46].

A few systems close the loop. CovidAlert buzzes on detected face touches [66]. In a case report on HabitAware's vibrating bracelet, one participant found false alarms rare, but a second, who used it for skin-picking, stopped wearing it because everyday movements triggered too many [42]. WatchGuardian lets users define habits from a few examples and nudges them by vibration [41]. Its trigger required three consecutive positive windows, its schedule was partly fixed by a five-minute cooldown and a forced nudge after twenty silent minutes, and its false-nudge rate during the intervention is not reported. WatchHAR names a nail-biting nudge as an application but does not build it [78]. None of these systems states or controls how often a wearer will be buzzed by mistake.

2.2 Evaluating Detectors for Continuous Use

Activity recognition is usually scored per window or per segment under leave-one-subject-out cross-validation [14, 20, 32, 70]. Ward et al. showed that frame-level scores hide the event-level errors that matter in continuous use, such as fragmented and merged events and timing shifts [75]; sound event detection makes a similar distinction [51]. Cherian et al. studied the NULL class in the wild [18], and Ding et al. argue that classification metrics do not fit continuous wearable monitoring and flag alarm fatigue as a risk [21]. Some communities already report false alarms per unit time: seizure detection per day [30, 34, 50, 56], real-world fall detection per hour or day [7, 57], and RisQ

false smoking sessions per day [58]. We bring this convention to activity-triggered nudges and test, across detectors, datasets and trigger policies, how well window metrics predict the per-wearer rate. Day-long, camera-annotated wrist datasets such as CAPTURE-24 [16] make such tests possible over whole days of everyday life.

2.3 False Alarms, Trust and Timing

False alarms make people respond less, the cry-wolf effect [11, 13]. Human-factors work separates the effects of false-alarm-prone and miss-prone automation on compliance and reliance [22, 52, 63], and places the usefulness of imperfect diagnostic aids at around 70% reliability [77]. Clinical alarm fatigue is the high-stakes version of the problem [19], and reliability shapes trust in haptic warnings [24] as it does trust in automation generally [33, 40]. This literature concerns operators who monitor an external process. In activity-triggered nudging, the wearer is the one being monitored, and the buzz often arrives during the movement it is about. Just-in-time adaptive interventions [36, 38, 55] and work on interruptibility [49, 59, 60] optimise *when* to intervene. TrueTap is complementary: it bounds how often an intervention is *wrong*.

2.4 Distribution-Free Guarantees

Temperature scaling [31] and selective classification [28] improve the meaning of model scores but give no rate guarantee. Conformal prediction gives finite-sample guarantees under exchangeability [2, 73], and extends to risk control [3, 4, 9], Neyman–Pearson control of type-I error [72], hierarchical exchangeability [23] and online settings [26, 29]. Recent work controls false-positive rates in event streams [45, 79] and studies uncertainty in activity recognition [8, 15, 64, 65]. Garcia-Ceja et al. found that calibrating on other users broke conformal coverage on one of their activity datasets, and that calibrating on the target user restored it [27].

BCT is an instance of this machinery: split conformal risk control with a user-level 0–1 loss [4], with users as the exchangeable unit [23]. Learn-then-Test and the Neyman–Pearson umbrella algorithm choose a larger order statistic so that the statement holds for the shipped calibration set [3, 72]; we also evaluate the most conservative choice, the largest order statistic. We do not claim new statistics. Our contribution is the quantity and what it takes to certify it: false buzzes per wearer-hour *after* debouncing, re-arming and cooldowns, a loss that is not monotone in the threshold and is only as fine as the exposure it is measured over; and an evaluation on real detectors that shows where every conventional rule fails, where the certificate holds, and where it breaks.

3 Problem Formulation

3.1 From Window Scores to Buzzes

An activity detector maps each window of sensor data to a score $s_t \in [0, 1]$ for the target behaviour. A *trigger policy* turns the score stream into nudges. TrueTap's policy smooths the score with a causal mean $\bar{s}_t$ over the last k windows and fires when $\bar{s}_t \geq \tau$, provided that the trigger is *armed*, that is, $\bar{s}$ has dropped below τ since the last nudge, and that at least R seconds have passed since the last nudge. Re-arming makes one sustained behaviour produce one nudge, and the refractory period stops flicker around the threshold from producing a burst. What the wearer feels depends on the whole policy (k, τ, R), not only on the detector.

3.2 Wearer-Centred Quantities

A target *episode* $e = [a_e, b_e]$ is a maximal run of target-labelled time, after merging runs separated by less than a task-specific gap. Some time is *don't-care*: episodes shorter than a task-specific minimum, unlabelled time and, where stated, sleep or ambiguous labels. A nudge at time t is *true* if

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

$a_e \le t \le b_e + \Delta$ for some episode, with tolerance $\Delta = 5$ s because a causal detector must lag onset; it is *ignored* if the same holds for a don't-care span, and *false* otherwise. A wearer's *non-target exposure* H^- is the labelled, non-don't-care time outside every $[a_e, b_e + \Delta]$, in hours. For each wearer we report **false nudges per hour**, FNH = #false$/H^-$, following the per-time convention of seizure and fall detection [56, 57]; **event recall**, the fraction of episodes with at least one true nudge; and **onset latency**, the time from a_e to the first true nudge.

Two clocks matter. FNH is a rate per hour of non-target exposure, but a wearer lives in wall-clock hours, which include target and don't-care time. The two differ by the non-target share $\phi = H^-/H_{\text{wall}}$. On CAPTURE-24, where sleep and unlabelled time are don't-care, the median participant's ϕ for eating is 0.35, so a budget of B per exposure hour allows about $0.35\,B$ false nudges per wall-clock hour. FNH is also only as fine as its exposure: with H^- hours it moves in steps of $1/H^-$, so a budget of one per hour certified on 45-minute sessions means "no false nudge in the session". Calibration data must be long enough to resolve the budget being promised.

3.3 The Statistical Target

We want a threshold $\hat{\tau}$, chosen *before* meeting a wearer, such that

$$\Pr\left(\text{FNH}_{\text{new}}(\hat{\tau}) \le B\right) \;\ge\; 1 - \delta$$

for a new wearer drawn from the same population as the calibration users, over a session like theirs. Users, not windows, are the unit. Windows from one person are strongly dependent, a per-window error rate says nothing about how often one person is buzzed, and window-level conformal coverage calibrated on other users can fail for a new user [27].

A new wearer needs care. Each CAPTURE-24 participant contributes one recorded day, and in the other datasets each person contributes one recorded session. The statement is therefore about a new *person-session*: a person we have not seen, observed over a session like the calibration sessions. It says nothing about the same person on other days, although a product sits on one wrist for months, and Section 7.3 shows that violations concentrate in a few people. Section 4.2 states exactly what the probability covers.

4 TrueTap

TrueTap sits between an activity detector and the vibration motor (Figure 2). It does not replace the classifier. It chooses the firing threshold of the trigger policy from users other than the wearer, so that the share of new wearers who exceed a false-nudge budget is controlled.

4.1 Budget-Calibrated Triggering

BCT takes a detector and a trigger policy (k, R), both fixed on training users, and n disjoint calibration users.

(1) For each calibration user i, compute $\text{FNH}_i(\tau)$ over a finite threshold grid $\mathcal{G} \subset [0.5, 1]$ that ends in a "never fire" point.

(2) The false-nudge count is not monotone in τ, for two reasons. Raising the threshold can split one excursion of the score into two crossings. And because the refractory period follows every nudge, a true nudge can block a false one that comes within R seconds; raising τ can remove the true nudge and let the false one through. Define each user's *certified threshold* through a monotone envelope, which covers both,

$$\tau_i^\star = \min\{\tau \in \mathcal{G} :\ \text{FNH}_i(\tau') \le B \ \ \forall\, \tau' \in \mathcal{G},\ \tau' \ge \tau\}.$$

The set is never empty, because the never-fire point has FNH = 0.

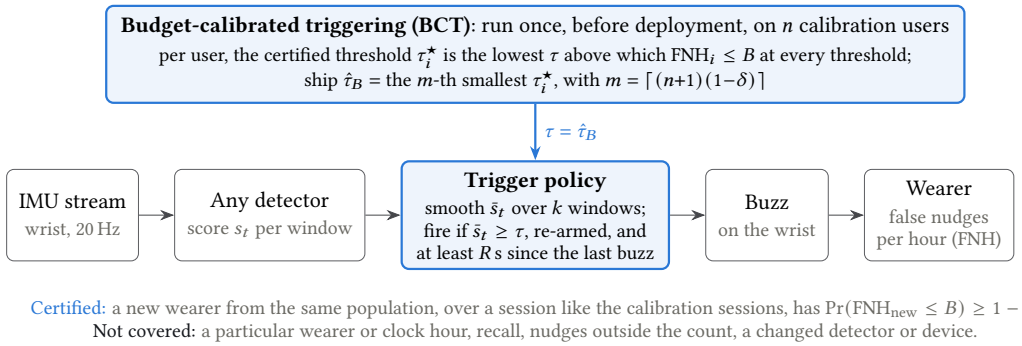


Fig. 2. TrueTap. Any detector scores each window of motion data; a trigger policy with smoothing, re-arming and a refractory period turns the scores into buzzes. Budget-calibrated triggering (BCT, blue) chooses the trigger's threshold once, before deployment, from calibration users other than the wearer. The certificate covers the share of new wearers whose false nudges per hour exceed the budget, and nothing else (Section 4.2, Table 1).

(3) Return the m-th smallest $\tau_i^\star$, with $m = \lceil (n+1)(1-\delta) \rceil$. If $m > n$ (for $\delta = 0.1$, whenever $n < 9$), return the never-fire point and report that the budget cannot be certified with n users.

We also report the variant $m = n$, which takes the most conservative calibration user, and a *naive quantile* without the +1 correction, as a check that the correction matters.

Figure 3 runs these steps on real data: the 50 calibration participants of one CAPTURE-24 split, for eating at $B = 1$/h. For most participants, false nudges fall as the threshold rises, and 31 are already within budget at the lowest threshold. The highlighted participant shows why the envelope is needed: raising τ from 0.53 to 0.54 adds a false nudge (3 instead of 2 over the day, 1.1 instead of 0.8 per hour), so their certified threshold is 0.55, although they were within budget at 0.50. BCT ships the 46-th smallest certified threshold, $\hat{\tau}_B = 0.60$. Of the split's 51 test participants, 5 exceeded the budget at that threshold.

The grid starts at 0.5 for a reason that lies in the trigger, not the classifier. At a very low threshold the smoothed score rarely drops below τ, so the trigger rarely re-arms, and false nudges and recall fall together for reasons unrelated to the detector. Calibration would happily select such thresholds. We observed this on HARTH, where without a floor BCT's threshold for running at $B = 4$/h fell below 0.1 in 16 of 20 splits, and set the floor to the conventional 0.5 rather than searching for a value. Any grid fixed in advance leaves the guarantee intact. Because we chose the floor after seeing the WISDM and HARTH results, only the CAPTURE-24 analysis, run afterwards, is strictly prospective; without any floor, BCT also stayed within its bound on WISDM (8.1%) and HARTH (6.2%).

4.2 What Is and Is Not Guaranteed

Proposition 4.1 (Marginal false-nudge budget). *Suppose the calibration users and a new user are exchangeable, given the training users. The detector and trigger policy are fixed by the training users alone, and $\mathcal{G}$ is chosen without reference to calibration or test users. Then, for $m \le n$, the BCT threshold satisfies*

$$\Pr\left(\mathrm{FNH}_{new}(\hat{\tau}) \le B\right) \;\ge\; \frac{m}{n+1} \;\ge\; 1 - \delta,$$

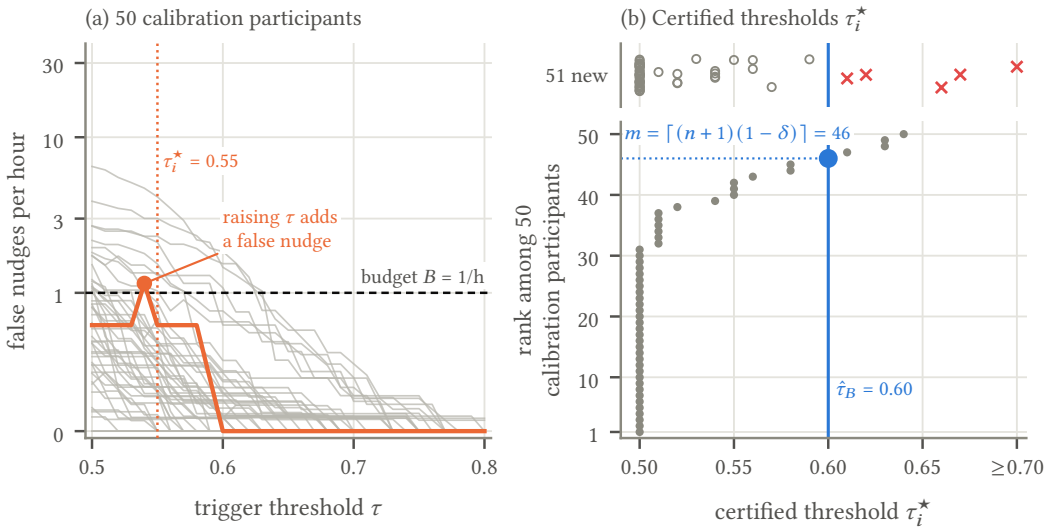


Fig. 3. BCT on real calibration data (CAPTURE-24, eating, $B = 1$ false nudge per exposure hour, $\delta = 0.1$, first split). (a) False nudges per hour against the trigger threshold for each of the 50 calibration participants. The highlighted participant gains a false nudge when τ rises from 0.53 to 0.54, so their certified threshold $\tau_i^\star$, above which they stay within budget at every threshold, is 0.55. (b) Bottom: the 50 certified thresholds, sorted; BCT ships the m-th smallest, $\hat{\tau}_B = 0.60$. Top: the certified thresholds of the 51 test participants, whom calibration never saw; the 5 marked × exceeded the budget at $\hat{\tau}_B$. Values of 0.70 or more are drawn at 0.70.

where the probability is over the draw of calibration users and *the new user. The violation probability is thus at most* $1 - m/(n+1)$*: 9.5% for* $n = 20$*, 7.7% for* $n = 12$ *and 9.8% for* $n = 50$ *at* $\delta = 0.1$*. For* $m = n$ *it is at most* $1/(n+1)$.

The proof (Appendix A) is the standard split-conformal rank argument; ties among the $\tau_i^\star$, which a finite grid makes common, only make it conservative. In statistical terms, BCT is split conformal risk control [4] with the user-level loss $\ell_i(\tau) = \mathbf{1}\{\mathrm{FNH}_i(\tau) > B\}$, and the envelope is the standard monotonisation of a non-monotone loss. Conditional on one calibration set, BCT's own violation rate is random: without ties it is at most a $\mathrm{Beta}(n+1-m, m)$ variable. The $m = n$ variant gives $\Pr_{\mathrm{cal}}(\text{violation} > \delta) \le (1-\delta)^n$, at most 0.1 once $n \ge 22$, which is the training-conditional statement that Learn-then-Test and the Neyman–Pearson umbrella algorithm target [3, 72].

Table 1 separates what is certified from what is only measured. A certificate on FNH alone can always be met by silence, so recall and latency are reported next to every result.

5 Evaluation Setup

5.1 Datasets and Tasks

We use four public datasets that play different roles (Table 2; integrity checks in Appendix B).

CAPTURE-24 [16] is our main test of everyday wrist use: 151 adults wore a wrist accelerometer (Axivity AX3, dominant wrist) for about 24 hours of free living. Activities were annotated from wearable-camera images with Compendium of Physical Activities codes, and sleep from diaries. We define four nudge tasks: *eating*, for mindful-eating prompts; *in vehicle*, for a driving or do-not-disturb mode; *bicycling*, for ride detection; and *sports*, for workout detection. Classes follow the release's "WillettsSpecific2018" daily-living scheme, with an added *eating* class for the Compendium eating codes, and recordings were reduced from 100 to 20 Hz by averaging blocks of five samples.

Table 1. What TrueTap guarantees, and what it only measures.

Component or situation	Status	What holds
BCT; exchangeable users; grid and policy fixed in advance; sessions like the calibration sessions	**Certified** (finite-sample, marginal)	$\Pr(\mathrm{FNH}_{\mathrm{new}} \le B) \ge m/(n+1)$, over calibration draw and new user (Proposition 4.1)
BCT with $m = n$	**Certified**, training-conditional for $n \ge 22$	$\Pr_{\mathrm{cal}}(\text{violation rate} > \delta) \le (1-\delta)^n$
A particular wearer, or the same wearer on other days	Not guaranteed	Some wearers exceed the budget under almost every calibration set (Section 7.3)
A particular clock hour	Not guaranteed	FNH averages over the session; hourly bursts are measured (Section 7.4)
Recall and onset latency	Not controlled	Reported; a budget can cost most of the recall (Sections 6.2 and 7.7)
Nudges in don't-care time	Not counted	Felt by the wearer but outside FNH (Section 7.5)
New device, detector or population	Not covered	Requires recalibration; score drift breaks the guarantee (Section 7.6)
Several nudge types on one watch	Per type	All q budgets hold together with probability at least $1 - q\delta$ (union bound) unless calibrated jointly

Exposure is waking, labelled, non-target time. Unlabelled time and sleep are don't-care, and so, for eating, are the desk and sitting codes annotated "with or without eating".

WISDM-2019 [76] is a controlled stress test with look-alikes: a smartwatch recorded 51 people, each doing nominally the same 18 scripted activities for about three minutes; we resampled all recordings to 20 Hz (Appendix B). Its four tasks (eating, brushing teeth, drinking from a cup and jogging) each have built-in look-alikes. Because every user does nominally the same activities for about the same time, users differ mainly in how they move; this is a best case for exchangeability, not a test of realism. Each user's bouts form one session with 0.59–0.85 h of exposure, so one false nudge is worth 1.2–1.7 FNH: $B = 1$/h means *zero* false nudges in the session, $B = 2$/h at most one, and $B = 4$/h two or three.

HARTH [44] gives free-living streams from 31 people (median 1.4 h each) from accelerometers on the lower back and thigh rather than the wrist; its tasks are running and cycling. **Daphnet** [6] is a small clinical population of 10 people with Parkinson's disease, with sensors on the ankle, thigh and trunk; its task, freezing of gait, stands in for vibrotactile cueing (the original system cued by sound).

5.2 Detectors and Trigger Policy

Detectors are multi-class, as in HAR practice, and a task's score is the summed probability of its target classes. For the calibration experiments, the detector is LightGBM [35] on 19 hand-crafted features per channel, trained afresh in every split on that split's training users. On CAPTURE-24, trained on 50 people, it reached a window accuracy of 64% over 11 classes, and a window F1 of 0.22 for eating, 0.53 for in vehicle, 0.74 for bicycling and 0.11 for sports. Daily-living recognition on this dataset is hard in general: the data paper's own baselines reach a macro F1 of 0.39–0.53 on its daily-living labels [16]. For the comparison of window metrics with the burden (Section 6.1), WISDM,

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

Table 2. Datasets and the role each plays. Exposure is the non-target time per participant and task over which FNH is computed.

Dataset	People	Sensor	Exposure per person	Role
CAPTURE-24	151	dominant wrist, acc.	median 9.0–10.9 h (waking)	everyday wrist use, about a day each
WISDM-2019	51	dominant wrist, acc. + gyro.	0.59–0.85 h	scripted look-alikes; best case for exchangeability
HARTH	31	lower back + thigh, acc.	median 1.1–1.4 h	free-living, not wrist
Daphnet	10	ankle, thigh, trunk, acc.	8–37 min	small clinical population

HARTH and Daphnet each have 14 detectors from five families (supplementary Section S2.2), scored by 5-fold cross-validation grouped by user.

Windows are 2 s with a 1 s hop on WISDM, HARTH and Daphnet, and 5 s without overlap on CAPTURE-24, which keeps day-long recordings for 151 people tractable. The trigger uses $k = 3$ and $R = 60$ s (10 s for freezing of gait), with tolerance $\Delta = 5$ s. On HARTH and CAPTURE-24, episodes are merged across gaps under 30 s, and episodes shorter than 30 s are don't-care. The pipeline is implemented in a Python library. A browser runtime of the WISDM detector and the trigger matches the library on held-out windows; it has been run only in Node, not on a phone or watch (supplementary Section S2.3).

5.3 Splits, Rules and Statistics

Training, calibration and test users are disjoint in every split: 10 random splits of 50/50/51 users on CAPTURE-24, 20 splits of 20/20/11 on WISDM and 20 of 12/12/7 on HARTH. Daphnet uses nested leave-one-patient-out (Section 7.7). For each task, budget $B \in \{1, 2, 4\}$ per hour and risk $\delta \in \{0.1, 0.2\}$, we compare on the test users a fixed $\tau = 0.5$; the window-F1-optimal threshold on calibration users' windows; the *median-user* threshold, the median of the calibration users' certified thresholds; the *pooled-rate* threshold, the lowest τ whose FNH pooled over calibration users is within B, which controls the average rather than the tail; the naive quantile; BCT; and BCT with $m = n$. Two references complete the comparison: an in-sample per-user *oracle* that knows each test user's curve and maximises recall within budget, and a sweep of every fixed threshold on the test users, which traces the whole trade-off (Section 6.3).

Evaluations are not independent: a person appears in several splits, and all tasks and budgets in a split share one detector and one calibration set. We therefore report 95% intervals from a two-way cluster bootstrap with 400 replicates, which resamples people and splits and keeps every evaluation of a sampled person in a sampled split. With 10 or 20 splits, the calibration-draw part of the variation is estimated coarsely. Pooled numbers count test-user evaluations, which come from 149 distinct CAPTURE-24 participants (those who were tested in at least one split), 51 WISDM users, 31 HARTH users and 10 Daphnet patients.

6 Results

We first ask how well window metrics predict the per-wearer burden (Section 6.1), then whether calibrated thresholds keep new wearers within budget and at what cost (Section 6.2), and finally what any global threshold could have achieved instead (Section 6.3).

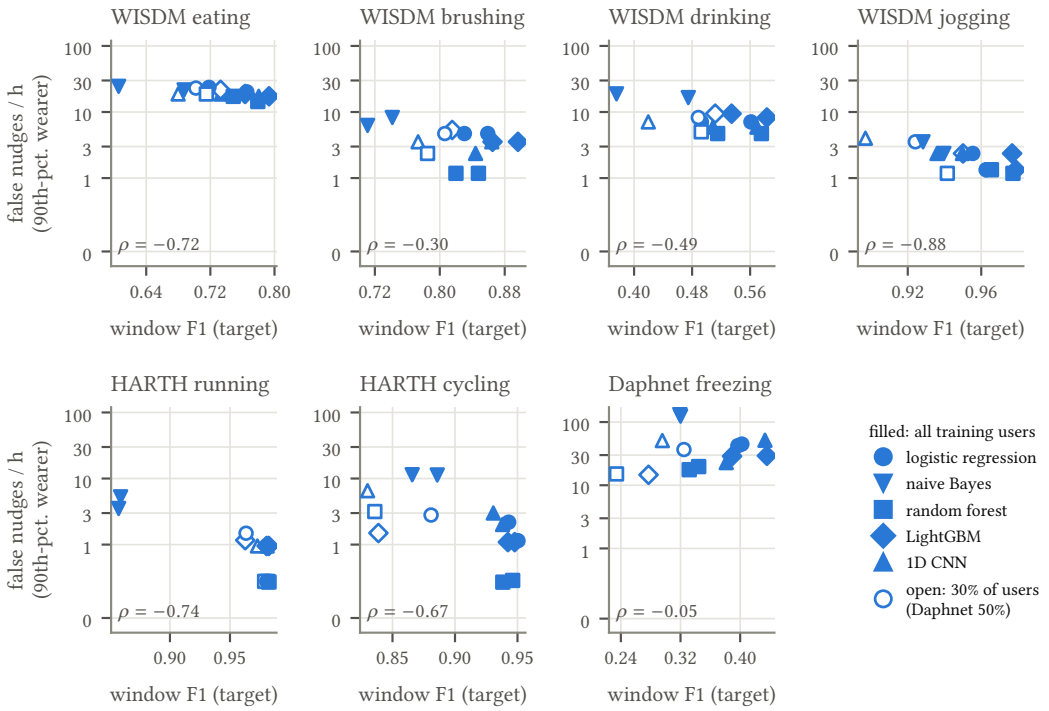


Fig. 4. Window F1 for the target behaviour against the false nudges per hour received by the 90th-percentile wearer, at the conventional operating point ($\tau = 0.5$, $k = 3$, task-default R). Filled markers: the 10 detectors per task trained on all training users (primary analysis). Open markers: deliberately weakened detectors trained on 30% of users (50% for Daphnet; sensitivity analysis). Spearman ρ in each panel is computed across the filled detectors. The vertical axis is linear below 1/h and logarithmic above; horizontal ranges differ between panels.

6.1 Window Metrics and the Per-Wearer Burden

Window metrics rank detectors only loosely by the burden. Over the 70 detector–task points from the 10 full-data detectors on seven tasks (Figure 4), within-task ranks of window F1 correlated with the 90th-percentile wearer's FNH at $\rho = -0.55$, and multi-class accuracy at $\rho = -0.67$. Window F1 is the binary F1 of the target class under the arg-max decision. To get intervals, we bootstrapped people and recomputed both the window metrics and the wearers' FNH. Resampling duplicates people, which pulls the bootstrap correlations towards zero (median −0.49 for F1), so we report both the plain percentile intervals (F1 −0.65 to −0.31, accuracy −0.69 to −0.47) and bias-corrected ones (F1 −0.73 to −0.43, accuracy −0.77 to −0.64), which rest on a large correction (supplementary Section S3.2). Either way, the correlation is clearly negative and far from −1. Within single tasks, ρ ranged from −0.88 to −0.05 and was weakest for freezing of gait, and adding the four weakened detectors per task changed little ($\rho = -0.51$, 98 points). Window metrics carry real signal about the burden, but detectors with similar F1 can differ several-fold in it.

Window metrics do not show the level of the burden. With the deployable detector at the conventional operating point, 100% of WISDM users received more than one false eating nudge per hour (median 11.8/h), and every user was buzzed by mistake at least once in their session, even though event recall was 0.97. For brushing teeth the median user received no false nudges, but the

Table 3. Share of *new* test participants whose false nudges per hour exceeded the budget, pooled over tasks and budgets $B \in \{1, 2, 4\}$/h at $\delta = 0.1$, with two-way (participant × split) cluster-bootstrap 95% intervals; mean event recall over participants with at least one target episode; and median onset latency. "Bound" is the finite-sample upper bound on the share over budget from Proposition 4.1, which holds for each task and budget separately. Pooled figures hide large differences between tasks (Figure 5).

Threshold rule ($\delta = 0.1$)	Over budget [95% CI]	Bound	Recall	Latency
CAPTURE-24 (wrist, free-living; 151 people); $n_{\text{cal}} = 50$				
Fixed $\tau = 0.5$	7.6% [5.4, 10.3]		0.65	75 s
Window-F1-optimal[a]	31.5% [28.8, 35.0]		0.83	52 s
Median user	7.5% [5.3, 10.0]		0.65	75 s
Pooled rate	7.2% [5.2, 9.6]		0.64	76 s
Naive quantile (no +1)	4.1% [2.6, 6.1]	11.8%	0.61	81 s
BCT	3.7% [2.2, 5.5]	9.8%	0.60	82 s
BCT, $m = n$	0.9% [0.4, 1.7]	2.0%	0.53	92 s
Per-user oracle[b]	0.0%		0.65	68 s
WISDM (wrist, scripted; 51 people); $n_{\text{cal}} = 20$				
Fixed $\tau = 0.5$	55.7% [52.6, 58.8]		0.96	2 s
Window-F1-optimal[a]	58.6% [55.1, 62.6]		0.97	2 s
Median user	36.5% [31.4, 41.9]		0.93	3 s
Pooled rate	30.9% [26.0, 35.6]		0.92	3 s
Naive quantile (no +1)	11.6% [8.6, 15.1]	14.3%	0.85	5 s
BCT	7.4% [5.2, 9.6]	9.5%	0.82	5 s
BCT, $m = n$	3.0% [1.7, 4.6]	4.8%	0.77	6 s
Per-user oracle[b]	0.0%		0.90	2 s
HARTH (back + thigh, free-living; 31 people); $n_{\text{cal}} = 12$				
Fixed $\tau = 0.5$	6.1% [1.9, 11.2]		0.99	3 s
Window-F1-optimal[a]	24.2% [16.5, 33.3]		0.95	2 s
Median user	6.1% [1.6, 12.2]		0.99	3 s
Pooled rate	6.1% [1.7, 12.3]		0.99	3 s
Naive quantile (no +1)	5.1% [1.1, 10.7]	15.4%	0.99	3 s
BCT	2.7% [0.3, 7.3]	7.7%	0.99	4 s
BCT, $m = n$	2.7% [0.3, 7.4]	7.7%	0.99	4 s
Per-user oracle[b]	0.0%		1.00	3 s

[a] Chosen on the full grid, so it can fall below 0.5; all other rules choose $\tau \geq 0.5$. [b] In-sample on each test user; within budget by construction. Bound: finite-sample upper bound on the share over budget. On HARTH ($n = 12$) BCT already has $m = n$.

90th-percentile user received 3.5/h. The Daphnet detector of the introduction shows the same gap in its starkest form. Figure S1 in supplementary Section S3.1 shows every wearer's FNH.

On scripted wrist tasks, the trigger policy matters as much as the detector. Decomposing the variance of log(1+pooled FNH) across the full-data detectors and nine trigger policies ($k \in \{1, 3, 5\}$, $R \in \{10, 60, 300\}$ s), the policy explained a median 51% and the detector 44%. The policy explained more than the detector on all four of WISDM's scripted wrist tasks (51–95%), and the detector more on HARTH (72–89%). The split depends on the ranges chosen, but smoothing and refractory settings can move the wearer's experience as much as the choice of classifier.

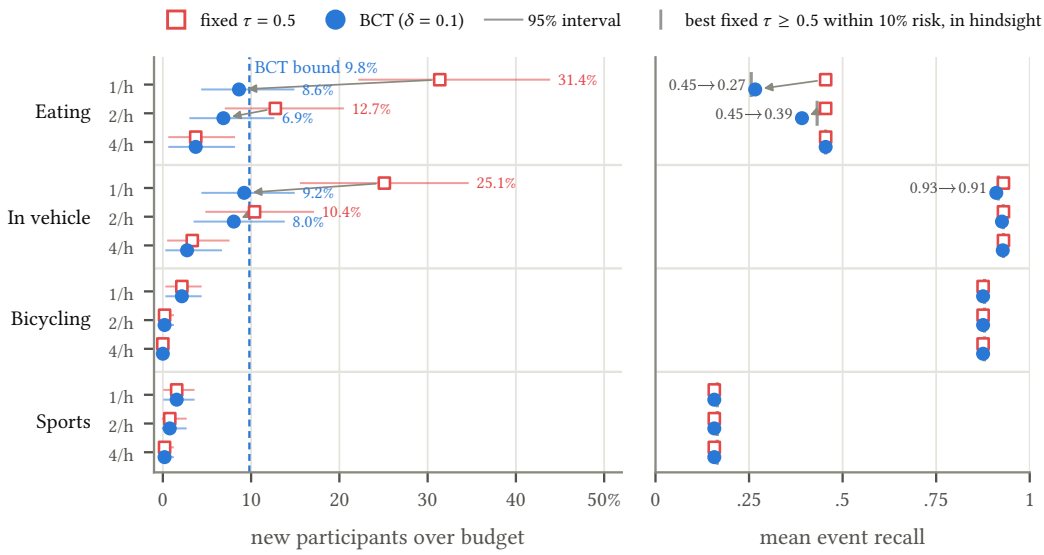


Fig. 5. CAPTURE-24 by task and budget ($\delta = 0.1$; 10 splits, 51 new test participants each). Left: share of new participants over budget for the conventional threshold (squares) and BCT (circles), with two-way cluster-bootstrap 95% intervals; arrows show what calibration changed. BCT's finite-sample bound is 9.8% in every row. Right: mean event recall for the same two rules, and the best fixed threshold $\tau \geq 0.5$ that kept at most 10% of the pooled test participants over budget, chosen in hindsight (grey tick, often hidden under the markers). Calibration helped where the conventional threshold failed (eating and in vehicle at 1 and 2/h) and left the other cells unchanged. Its cost was small for in-vehicle detection and large for eating, where the hindsight threshold did no better. Appendix C gives every value with latencies.

6.2 Keeping New Wearers Within Budget

Pooled over tasks, conventional thresholds look acceptable; per task, they do not. Pooled over CAPTURE-24's four tasks and three budgets, the fixed threshold $\tau = 0.5$ left 7.6% of new participants over budget (95% interval 5.4–10.3%; Table 3), the median-user threshold 7.5% and the pooled-rate threshold 7.2%. These figures sit below $\delta = 0.1$ only because two tasks were easy (Figure 5). Across the 12 task–budget combinations, the fixed threshold's share ranged from 0.0% to 31.4% and exceeded 10% in 4: eating and in-vehicle detection at 1 and 2/h. The median-user and pooled-rate rules failed in the same 4 combinations (worst 31.0% and 28.2%), because keeping the *average* calibration participant within budget says nothing about the tail. The window-F1-optimal threshold, which may fall below 0.5, was worse still (31.5% pooled). The other datasets show the same pattern: on WISDM's scripted look-alikes the fixed threshold left 55.7% over budget, from 0.5% for jogging at 4/h to 100.0% for eating at 1/h; on HARTH, 6.1% overall but 17.1% for cycling at 1/h.

BCT stayed near or below its finite-sample bound. At $\delta = 0.1$, BCT left 3.7% of new CAPTURE-24 participants over budget (95% interval 2.2–5.5%), against a bound of 9.8% for $n = 50$ calibration participants, and its point estimate was below 10% in all 12 combinations. The highest were in-vehicle detection and eating at 1/h (9.2% and 8.6%), whose 95% intervals both reach 14.8%. The naive quantile left 4.1% over budget and the $m = n$ variant 0.9% (bound 2.0%); at $\delta = 0.2$, BCT left 5.7% (bound 19.6%). On WISDM, BCT left 7.4% over budget (bound 9.5%) and on HARTH 2.7% (bound 7.7%). No HARTH combination exceeded its bound; on WISDM, two of 12 did: jogging at 1/h (11.4%) and drinking at 4/h (9.55% against 9.52%), each over 220 evaluations from 51 people. A

marginal bound checked cell by cell is expected to be exceeded in a few cells by sampling error alone. The pooled CAPTURE-24 figures count 6,120 test-participant evaluations per rule and δ, from 149 distinct people with one recorded day each.

The cost is recall and latency, and it falls on the tasks that need the budget. Pooled over tasks, BCT's mean recall on CAPTURE-24 was 0.60, against 0.65 for the fixed threshold and 0.65 for the in-sample per-user oracle, but the pooled mean hides four different situations (Figure 5, right). For *eating*, the budget bound and cost recall: at $B = 1$/h, BCT caught 0.27 of meals, against 0.45 for the fixed threshold and 0.41 for the oracle, and its median onset latency rose from 101 to 143 s. At $B = 4$/h the budget no longer bound, and BCT caught 0.45. For *in-vehicle detection*, the budget bound cheaply: at 1/h BCT cut the share over budget from 25.1% to 9.2% at a recall of 0.91 against 0.93. HARTH cycling (17.1% to 4.3%, recall 0.98) and WISDM brushing (50.5% to 5.0%, recall 0.91 against 0.95) behaved the same way. For *bicycling*, the budget did not bind, and BCT kept the fixed threshold's operating point (recall 0.88). For *sports*, the detector (window F1 0.11) and the threshold floor limited recall: both rules caught 0.16 of workouts, and even the in-sample oracle restricted to $\tau \geq 0.5$ caught only 0.17.

A designer can see the need and the cost before deployment. Both are visible in the calibration participants alone. At the fixed threshold, 35.4% of calibration participants were over budget for eating at 1/h and 28.2% for in-vehicle detection. Averaged over splits, the calibration share was within 4.0 percentage points of the test share in all 12 combinations, and calibration participants' recall at $\hat{\tau}_B$ (eating 0.29, in vehicle 0.90 at 1/h) was within 0.05 of the test participants' for every task except sports, where only 3–11 participants per group and split recorded a workout. A designer has only one calibration set, and within single splits the gaps reached 17 points and 0.13 in recall. The need was nonetheless visible in every split: at 1/h, 26–52% of a split's calibration participants were over budget for eating and 16–60% for in-vehicle detection.

6.3 What a Budget Buys: The Recall–Risk Trade-Off

A calibrated threshold is still one global threshold. To see what any global threshold could have achieved, we evaluated every fixed $\tau \in [0.5, 1]$ on the test participants (Figure 6), and for each split found, in hindsight, the threshold with the highest recall among those that kept at most 10% of that split's own test participants over budget. This uses outcomes that a deployment never sees. Averaged over splits, BCT's recall was within 0.04 of this hindsight threshold's on every CAPTURE-24 task and budget, and within 0.07 on WISDM and HARTH. In the 6 of 120 CAPTURE-24 split cells where BCT's recall was higher, BCT had left more than 10% of that split's test participants over budget.

Some budgets are infeasible for these detectors. We call a task–budget combination feasible if some fixed threshold kept at most 10% of the pooled test participants over budget while catching at least 80% of episodes. On CAPTURE-24, 6 of 12 combinations were infeasible: every budget for eating and sports. For eating, the best threshold within the risk target caught 0.26 of meals at 1/h and 0.45 at 4/h; allowing thresholds below 0.5 raised this only to 0.55 at 4/h, and to 0.27–0.43 of workouts. On WISDM and HARTH, 5 of 18 combinations were infeasible. In these combinations no threshold rule, calibrated or not, can deliver both the budget and useful recall; only a better detector, a different trigger policy or a looser budget can. BCT does not move the trade-off curve. It picks a point on it with a stated risk, before deployment, and makes the cost of that point visible.

7 Where the Guarantee Breaks

Proposition 4.1 rests on assumptions that deployments can violate, and on a definition of FNH that a wearer does not share. We probed each on CAPTURE-24, and score drift also on WISDM (first

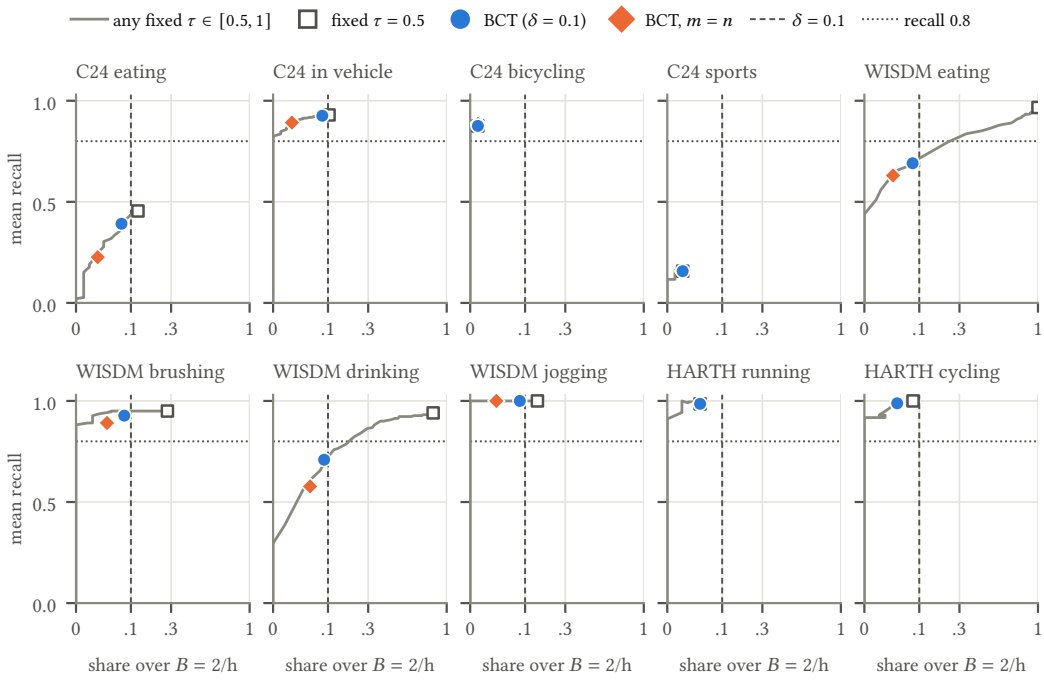


Fig. 6. Recall against the share of new participants over budget ($B = 2$/h) for every fixed threshold $\tau \in [0.5, 1]$ (grey curve, evaluated in hindsight on the pooled test participants of all splits), the conventional $\tau = 0.5$ (open square), BCT at $\delta = 0.1$ (blue circle) and its $m = n$ variant (orange diamond). C24: CAPTURE-24. The horizontal axis has a square-root scale. Points left of the dashed line meet the risk target; points above the dotted line catch at least 80% of episodes. BCT is chosen on calibration participants only, yet lands on or near the hindsight curve; it cannot move the curve. Where BCT coincides with the fixed threshold (C24 bicycling and sports, HARTH running), it hides the square.

ten splits, all budgets, $\delta = 0.1$; baseline 7.2% over budget at a recall of 0.83). Table 4 summarises the results; the subsections give what the table cannot. Supplementary Section S3.6 reports two further checks, on session order and on a published trigger policy.

7.1 Calibrating on One Group, Deploying on Another

Random splits make calibration and test users exchangeable by construction. As a real, if mild, shift, we used CAPTURE-24's age band (18–37 against 38 and over) and sex. In each split we calibrated BCT on 25 people of one group, limited by the smallest group, and tested it on the other group and on held-out people of the same group, over five draws per split, all splits, tasks, three budgets and four calibration groups (bound 7.7%). Across groups, 3.0% of test participants were over budget, against 3.1% within groups, and 4 and 3 of the 48 group–task–budget combinations exceeded the bound (at most 8.6% and 8.5%). The budget transferred about as well as it held within a group. Larger shifts, such as another device, wrist or country, remain untested.

7.2 The Threshold Floor

We recomputed BCT with grid floors of 0, 0.3, 0.5 and 0.7 (all values in supplementary Section S3.8). On WISDM the floor barely mattered. On HARTH, without it 53% of thresholds fell below 0.5

Table 4. Stress tests of the false-nudge budget on CAPTURE-24 (BCT, $\delta = 0.1$, bound 9.8% unless stated). "Over budget" is the share of new test participants whose FNH exceeded B. Each row gets one of three verdicts. *Holds*: the share is within the bound. *Not covered*: the probe asks about something the certificate does not state, such as one calibration set, one person or one clock hour. *Breaks*: the share exceeds the bound. The recall column shows what holding cost.

Probe (section)	What changes	Over budget	Recall	Verdict
Reference	Random splits, $n = 50$	3.7%	0.60	holds
Group shift (7.1)	Calibrate on one age band or sex, deploy on the other ($n = 25$; bound 7.7%)	3.0% across groups, 3.1% within	–	holds
Threshold floor (7.2)	Grid starts at 0, 0.3, 0.5 or 0.7	7.6, 5.1, 3.7, 0.5%	task-dependent	holds
One calibration set (7.3)	Only one split, as a designer would ship	1.3–7.2% across splits; $\Pr(\text{rate} > \delta) = 0.43$	–	not covered
Repeat violators (7.3)	Same person, different calibration sets	3% of cells give 51% of violations	–	not covered
Hourly bursts (7.4)	Count per clock hour, not per day	14% of within-budget days at 2/h had an hour with >2	–	not covered
Uncounted nudges (7.5)	Count unlabelled waking time too	4.6% overall; in vehicle at 1/h 16.7%, 9.2% after recalibrating; at 2/h 10.8% and 10.4% either way	in vehicle 0.91 → 0.81	breaks (in vehicle)
Score drift (7.6)	Scores shifted by 0.5, 1 or 2 logits	14.2, 33.6, 52.1%	–	breaks
Few calibration users (7.7)	Daphnet, $n = 9$ patients, $B = 2$/h (bound 10%)	1 of 10 patients	0.03 (oracle 0.24)	holds

and recall dropped from 0.99 to 0.93. On CAPTURE-24, BCT returned exactly 0.5 in 85 of 120 split–task–budget combinations, and the share over budget stayed within the bound at every floor. Removing the floor raised recall for sports (from 0.16 to 0.24–0.45 across budgets) and for eating at 4/h (from 0.45 to 0.54), and lowered it for bicycling (from 0.88 to 0.72–0.83), the re-arming artefact seen on HARTH. The floor trades recall on some tasks for robustness to the artefact on others, and a deployment could set it per task on separate users.

7.3 Personal Variation and One Calibration Set

The guarantee is marginal, and in practice that means two things. First, some wearers fail repeatedly. Among (user, task, budget) cells in which a user was tested at least three times, 4% on WISDM and HARTH exceeded the budget in at least half their splits, and they produced 44% of all BCT violations; on CAPTURE-24 the figures were 3% and 51%, where each repeat is the same recorded day tested against different calibration sets. Second, one calibration set can be unlucky. Given one calibration set, the violation rate exceeds $\delta = 0.1$ with probability 0.43 for CAPTURE-24 ($n = 50$) without ties, and ties can only lower this; across our ten splits BCT's share over budget ranged from 1.3% to 7.2%, and in 12 of 120 split–task–budget cells more than 10% of a split's 51 test participants were over budget (at most 17.6%, against 31 cells and at most 60.8% for the fixed threshold). The $m = n$ variant lowers that probability to 0.005, at the cost in recall shown in Table 3. A product that ships one calibration should use it. BCT is best seen as a population-level starting point, to be paired with a per-user correction such as an enrolment session [27].

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

7.4 Bursts Within a Day

A daily rate can hide bursts. We split each CAPTURE-24 test participant's recording into clock hours and kept the 21,609 participant–task hours with at least 30 minutes of exposure; the dropped hours hold 14% of the false nudges. At $B = 1$/h, 3.1% of these hours held more than one false nudge under BCT, against 7.3% under the fixed threshold. At $B = 2$/h the worst single hour held 31 false nudges under BCT, and among participant–task days that stayed within budget over the whole day, 14% still contained a clock hour with more than two. The certificate is a statement about the rate over the calibrated session, here a waking day. Controlling bursts would need an hourly loss in the calibration, at a further cost in recall.

7.5 Nudges Outside the Count

FNH counts only nudges during labelled, non-target waking time, but the wearer feels every nudge. At the BCT and fixed thresholds, 35% of all CAPTURE-24 nudges fell in don't-care time, almost all of it unlabelled or ambiguous *waking* time; at most 3.3% of a task's nudges fell during sleep. Counting every nudge, the median participant received 5 eating nudges per day under BCT at 1/h, against 16 under the fixed threshold. We therefore recomputed every participant's curve under a *strict* definition, in which only sleep and short target episodes remain don't-care, raising the median participant's exposure to about 18 h. This is a worst case, since some unlabelled time may contain the target or time when the watch was off the wrist. Under the strict count, BCT calibrated on the standard definition stayed within its bound overall but not for in-vehicle detection at 1/h (Table 4); recalibrating on the strict count restored it at a cost in recall, while at 2/h the share stayed slightly above the bound with either calibration (10.8% and 10.4%). The certificate is only as meaningful as the definition of a false nudge it is calibrated on, and a deployment should calibrate on the count its wearers will feel.

7.6 Score Drift

We shifted test users' scores in logit space, as a new device, a retrained model or a firmware change might, and kept the calibrated threshold. On CAPTURE-24, a one-logit shift raised the share over budget from 3.7% to 33.6%. On WISDM, shifts of 0.5, 1 and 2 logits raised it from 7.2% to 11.8%, 19.2% and 36.7%, and sharpening the scores (temperature 0.8) to 14.4%. BCT's thresholds lay mostly at the floor on CAPTURE-24 (between 0.50 and 0.76) and in the saturated tail of the score distribution on WISDM; in both cases a one-logit shift moved enough windows across them to break the budget for many more people. Any change of detector, firmware or device needs recalibration on a fresh, labelled cohort.

7.7 Few Calibration Users

Daphnet's ten patients are too few for random splits, so we used nested leave-one-patient-out: for each test patient, the other nine form the calibration set, and each of them is scored by a detector trained on the remaining eight, so that no score comes from a model that saw the test patient or the patient being scored. At $B = 2$/h the fixed threshold left 50% of patients over budget. With $n = 9$ and $\delta = 0.1$, BCT must take the most conservative patient's threshold: 1 of the ten patients exceeded the budget, too few to assess the risk, and recall fell to 0.03 (in-sample oracle 0.24). At $\delta = 0.2$, 3 exceeded it and recall was 0.12; ten patients cannot separate this from chance. With ten patients, an honest certificate leaves recall near zero, which for a cue meant to help someone through a freeze is the wrong trade.

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

8 Discussion

8.1 When a Budget Helps, When It Costs, and When It Fails

Two questions decide what a budget does for a task–budget combination: does it bind, that is, does the conventional threshold leave many new wearers over it; and can any threshold meet it while still catching most episodes? The answers belong to the combination, the detector and the population, not to the task alone (Figures 5 and 6).

Where the budget *does not bind*, as for bicycling and sports on CAPTURE-24, running on HARTH and jogging on WISDM at 4/h, calibration chose the conventional operating point or one close to it. The budget still helps, because it makes the promise checkable in advance, but it does nothing for recall. Where the budget *binds cheaply*, as for in-vehicle detection on CAPTURE-24, cycling on HARTH and brushing teeth on WISDM at 1/h, the conventional threshold left 17.1–50.5% of new wearers over budget, and calibration brought this near or below 10% for a few points of recall. Here calibration is most clearly worth having, and on CAPTURE-24 and HARTH a pooled evaluation would not have shown the need for it.

Where the budget *binds and costs recall*, as for eating on CAPTURE-24 at 1 and 2/h, eating on WISDM at every budget and drinking on WISDM at 1 and 2/h, the tasks are full of look-alikes. Calibration kept new wearers within budget at the price of missed episodes and later buzzes, and no global threshold within the risk target caught 80% of episodes with these detectors. On CAPTURE-24 we cannot separate the role of look-alikes from that of a weak eating detector (window F1 0.22). Whether an eating nudge that catches about a quarter of meals is worth shipping is a design decision that TrueTap does not make. It moves the decision earlier: the calibration participants alone showed the need and gave an estimate of the cost, and a designer can accept the cost, relax the budget or improve the detector. Finally, with *too few calibration users*, as for the ten Daphnet patients, an honest certificate left recall near zero. For a cue meant to help someone through a freeze, a false-nudge budget alone is the wrong objective.

8.2 What to Report

We recommend that papers on activity-triggered interventions report, next to window metrics, the median and 90th-percentile wearer's FNH, the exposure over which it was measured, event recall, onset latency, and the smoothing and refractory settings. Window metrics predicted the burden only moderately, and on scripted wrist tasks the trigger policy moved it as much as the detector did. Results should be reported per task and budget: on CAPTURE-24, pooling made a conventional threshold look acceptable when it failed for two of four tasks. Papers should also say which time counts as exposure and how many nudges fall outside it, since the wearer feels those too; on CAPTURE-24, the median participant's non-target share for eating was 0.35, so a budget of one per exposure hour is about 0.35 per wall-clock hour. Finally, short recordings resolve per-hour budgets only coarsely, so calibration data should be as long as the period the promise is meant to cover.

8.3 Deploying a Budget

BCT is a population-level starting point. Its guarantee is marginal over calibration sets and wearers, holds for sessions like the calibration sessions, and covers only the count it was calibrated on (Table 1). Section 7 showed what this leaves open, and each gap has a deployment cost. Any change of model, firmware or device needs recalibration, and on CAPTURE-24 that means a fresh cohort of camera-annotated days. The count should be the one wearers will feel, including time a dataset leaves unlabelled. A product that ships a single calibration should use the training-conditional $m = n$ variant and accept its recall cost. A watch that runs q nudge types, each calibrated at risk δ,

can only promise by the union bound that a new wearer stays within all of them with probability at least $1 - q\delta$, unless the budgets are calibrated jointly.

8.4 Open Problems

Each limit in Section 7 suggests a next step. *Bursts:* a loss on the worst clock hour rather than the session rate would bring bursts inside the promise, at a further cost in recall. *Per-wearer guarantees:* an enrolment session on the wearer's own wrist could turn the population statement into one about a person across days, as calibrating on the target user restored coverage in earlier activity-recognition work [27]. *Budget and recall together:* Learn-then-Test [3] can certify a recall or latency floor alongside the false-nudge budget, so that silence no longer satisfies the certificate; for freezing of gait, budgets would be chosen with clinicians. *Cheap recalibration:* because drift breaks the guarantee, recalibration that needs few new labels after a model, firmware or device change would decide whether budgets are practical at scale. *People:* budgets of 1, 2 and 4 per hour were chosen, not measured. A study with wearers should measure stated tolerance for false buzzes, dismissals, and whether people switch the feature off, and, for the confidence tiers in TrueTap's prototype (supplementary Section S1), whether wearers can feel the difference between a *sure* and a *maybe* buzz and act on it. Offline on CAPTURE-24 at $B \in \{2, 4\}$/h, *sure* nudges were correct 74.3% of the time and *maybe* nudges 49.2%.

8.5 Limitations

All results are offline, on recorded data with triggers replayed on recorded detector scores. No person has worn TrueTap or felt a budgeted nudge, and the runtime has not been run on a phone or watch. CAPTURE-24 labels come from wearable-camera images taken every 20–40 seconds and from sleep diaries [16], so our budgets are per hour of labelled, non-target waking time, onset noise inflates latency, and each participant contributes one day, which confounds day-to-day with person-to-person variation. We used one feature-based detector family for calibration, and the CAPTURE-24 detector is weak for some tasks; better detectors move the trade-off curve but do not change what the certificate means. The window-metric analysis does not include CAPTURE-24. Our day-long wrist results rest on one dataset of UK adults with one device, WISDM is scripted, HARTH is not wrist-worn, and Daphnet has ten patients.

9 Conclusion

A smartwatch that acts on what it infers should be able to say how often it will be wrong. We formulated that error as false nudges per hour for each wearer, after the full trigger policy, and showed that window metrics are only a moderate guide to it. Budget-calibrated triggering turns a population of calibration users into a checkable statement: a new wearer from the same population, over a session like theirs, exceeds the budget with probability at most δ. On day-long free-living wrist data, scripted look-alikes and non-wrist free-living data, the share of new wearers over budget stayed near or below its finite-sample bound, while conventional thresholds met the same budgets for some tasks and failed badly for others, and pooled reporting hid the failures. Where the budget bound, the cost fell on recall, and for some tasks no global threshold could meet both the budget and useful recall. The statement is narrow: it is marginal over wearers, covers a session-level rate of counted nudges, and breaks under modest score drift. Whether people use, trust or benefit from budgeted nudges remains to be tested on devices, with people. Until then, systems that act on their inferences should at least report their errors per wearer-hour, not per window.

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

Ethics

This work involved no research with human subjects: no one was recruited, contacted, observed or recorded, and no data were collected from people. It analyses four public, de-identified research datasets under their licences. A user study is planned but has not been run; it would need ethics approval before any data are collected.

Use of AI Tools

In line with ACM's policy on authorship, we disclose substantial use of generative AI. The authors chose the problem from candidate research gaps, led the problem formulation, the experimental methodology, the evaluation strategy and the interpretation of the findings, and made the final design and methodological decisions. Claude Opus 5.5 (Anthropic), used in October and November 2026 through the Claude app with a sandboxed environment for running code, searched the literature, including for the candidate research gaps; helped obtain and process the datasets; under the authors' direction, implemented and ran the Python library, the browser runtime and the analysis, figure and verification code; explored alternative analyses and identified limitations; and drafted the text of every section. The reruns of the pipeline recorded in the supplementary archive were also run in that environment. No image-generation model was used; all figures and tables come from scripts. The prompts Claude wrote for its helper agents are in the archive (`ai_use/`). The authors critically reviewed the AI-assisted outputs, and responsibility for the research, the reported findings and the conclusions rests with the authors.

All results come from scripts run on public datasets. The result numbers in the text, tables and figures, apart from the WISDM, Daphnet and HARTH integrity counts in Appendix B, are typeset from the result files by script, and a second script recomputes the key numbers from the raw per-user tables. No participant data were collected, and no result uses generated participant data: triggers are replayed offline on recorded detector scores, WISDM sessions join each user's recorded bouts in a seeded order, and the score perturbations of Section 7.6 are simulated, as stated where they are used.

Data and Code

We use CAPTURE-24 (CC BY 4.0) from its official archive, and WISDM-2019 (CC BY 4.0), Daphnet (CC BY 4.0) and HARTH under their published licences from public mirrors checked against published counts (Appendix B). The supplementary document (Sections S1–S3) and the archive with the code, all result files and the verification logs will be released publicly on acceptance.

References

[1] Abdullah Alesmaeil and Eftal Şehirli. 2025. Real-Time Nail-Biting Detection on a Smartwatch Using Three CNN Models Pipeline. *Computational Intelligence* 41, 1 (2025). doi:10.1111/coin.70020

[2] Anastasios N. Angelopoulos and Stephen Bates. 2021. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. arXiv:2107.07511 https://arxiv.org/abs/2107.07511

[3] Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. 2021. Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. arXiv:2110.01052 https://arxiv.org/abs/2110.01052

[4] Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. 2024. Conformal Risk Control. In *The Twelfth International Conference on Learning Representations (ICLR 2024)*. https://arxiv.org/abs/2208.02814

[5] Apple Inc. 2024. Set up Handwashing on Apple Watch (Apple Watch User Guide, watchOS 11). Apple Support documentation. https://support.apple.com/en-jo/guide/watch/set-up-handwashing-apdc9b9f04a8/11.0/watchos/11.0 Accessed 16 October 2026.

[6] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. Hausdorff, N. Giladi, and G. Tröster. 2010. Wearable Assistant for Parkinson's Disease Patients With the Freezing of Gait Symptom. *IEEE Transactions on Information Technology in Biomedicine* 14, 2 (2010), 436–446. doi:10.1109/TITB.2009.2036165

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

[7] Fabio Bagalà, Clemens Becker, Angelo Cappello, Lorenzo Chiari, Kamiar Aminian, Jeffrey M. Hausdorff, Wiebren Zijlstra, and Jochen Klenk. 2012. Evaluation of Accelerometer-Based Fall Detection Algorithms on Real-World Falls. *PLOS ONE* 7, 5 (2012), e37062. doi:10.1371/journal.pone.0037062

[8] Vladimir Balditsyn, Philippe Lalanda, German Vega, and Stéphanie Chollet. 2025. Quantifying Uncertainty in Machine Learning-Based Pervasive Systems: Application to Human Activity Recognition. arXiv:2512.09775 https://arxiv.org/abs/2512.09775

[9] Stephen Bates, Anastasios Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. 2021. Distribution-Free, Risk-Controlling Prediction Sets. arXiv:2101.02703 https://arxiv.org/abs/2101.02703

[10] Sarnab Bhattacharya, Rebecca Adaimi, and Edison Thomaz. 2022. Leveraging Sound and Wrist Motion to Detect Activities of Daily Living with Commodity Smartwatches. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 2, Article 42 (2022). doi:10.1145/3534582

[11] James P. Bliss, Richard D. Gilson, and John E. Deaton. 1995. Human Probability Matching Behaviour in Response to Alarms of Varying Reliability. *Ergonomics* 38, 11 (1995), 2300–2312. doi:10.1080/00140139508925269

[12] Bruce Brew, Steven G. Faux, and Elizabeth Blanchard. 2022. Effectiveness of a Smartwatch App in Detecting Induced Falls: Observational Study. *JMIR Formative Research* 6, 3 (2022), e30121. doi:10.2196/30121

[13] Shlomo Breznitz. 1984. *Cry Wolf: The Psychology of False Alarms.* Lawrence Erlbaum Associates, Hillsdale, NJ. ISBN 0898592968. https://ci.nii.ac.jp/ncid/BA01002810

[14] Andreas Bulling, Ulf Blanke, and Bernt Schiele. 2014. A Tutorial on Human Activity Recognition Using Body-worn Inertial Sensors. *Comput. Surveys* 46, 3 (2014), 1–33. doi:10.1145/2499621

[15] Maximilian Burzer, Till Riedel, Michael Beigl, and Tobias Röddiger. 2026. BayaHAR: Lightweight Bayesian Few-Shot User Adaptation for On-Device Personalized Human Activity Recognition. arXiv:2606.04798 doi:10.1145/3830727.3834833

[16] Shing Chan, Yuan Hang, Catherine Tong, Aidan Acquah, Abram Schonfeldt, Jonathan Gershuny, and Aiden Doherty. 2024. CAPTURE-24: A Large Dataset of Wrist-Worn Activity Tracker Data Collected in the Wild for Human Activity Recognition. *Scientific Data* 11 (2024), 1135. doi:10.1038/s41597-024-03960-3

[17] Xiang 'Anthony' Chen. 2020. FaceOff: Detecting Face Touching with a Wrist-Worn Accelerometer. arXiv:2008.01769 https://arxiv.org/abs/2008.01769

[18] Josh Cherian, Samantha Ray, Paul Taele, Jung In Koh, and Tracy Hammond. 2024. Exploring the Impact of the NULL Class on In-the-Wild Human Activity Recognition. *Sensors* 24, 12 (2024), 3898. doi:10.3390/s24123898

[19] Maria Cvach. 2012. Monitor Alarm Fatigue: An Integrative Review. *Biomedical Instrumentation & Technology* 46, 4 (2012). doi:10.2345/0899-8205-46.4.268

[20] Akbar Dehghani, Tristan Glatard, and Emad Shihab. 2019. Subject Cross Validation in Human Activity Recognition. arXiv:1904.02666 https://arxiv.org/abs/1904.02666

[21] Cheng Ding, Zhicheng Guo, Cynthia Rudin, Ran Xiao, Fadi B. Nahab, and Xiao Hu. 2023. Reconsideration on Evaluation of Machine Learning Models in Continuous Monitoring Using Wearables. arXiv:2312.02300 https://arxiv.org/abs/2312.02300

[22] Stephen R. Dixon, Christopher D. Wickens, and Jason S. McCarley. 2007. On the Independence of Compliance and Reliance: Are Automation False Alarms Worse Than Misses? *Human Factors* 49, 4 (2007), 564–572. https://apps.dtic.mil/sti/pdfs/ADA496817.pdf Technical-report version: University of Illinois AHFD-05-16/MAAD-05-04.

[23] Robin Dunn, Larry Wasserman, and Aaditya Ramdas. 2023. Distribution-Free Prediction Sets for Two-Layer Hierarchical Models. *J. Amer. Statist. Assoc.* 118, 544 (2023), 2491–2502. doi:10.1080/01621459.2022.2060112

[24] Mario J. Enriquez and Karon E. MacLean. 2004. Impact of Haptic Warning Signal Reliability in a Time-and-Safety-Critical Task. In *Proceedings of the 12th International Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (HAPTICS '04)*. IEEE. https://www.cs.ubc.ca/labs/spin/node/253

[25] Ramin Fathian, Steven Phan, Chester Ho, and Hossein Rouhani. 2022. A Wearable Sensor to Detect Face Touch and Prevent Spread of Infection. In *Proceedings of the Canadian Society for Mechanical Engineering International Congress 2022*. https://era.library.ualberta.ca/items/7cf3981d-57fc-48e7-a784-0a740d9d2370/download/c446bccb-660e-4775-87f3-0c77e87f98be

[26] Shai Feldman, Liran Ringel, Stephen Bates, and Yaniv Romano. 2023. Achieving Risk Control in Online Learning Settings. arXiv:2205.09095 https://arxiv.org/abs/2205.09095

[27] Enrique Garcia-Ceja, Luciano Garcia-Banuelos, and Nicolas Jourdan. 2023. Conformal Prediction in Multi-User Settings: An Evaluation. arXiv:2312.05195 https://arxiv.org/abs/2312.05195

[28] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*. https://arxiv.org/abs/1705.08500

[29] Isaac Gibbs and Emmanuel Candès. 2021. Adaptive Conformal Inference Under Distribution Shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*. 1660–1672. https://proceedings.neurips.cc/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

[30] D. M. Goldenholz, P. J. Karoly, P. F. Viana, E. Nurse, T. Loddenkemper, A. Schulze-Bonhage, S. Vieluf, E. Bruno, M. Nasseri, M. P. Richardson, B. H. Brinkmann, and M. B. Westover. 2024. Minimum clinical utility standards for wearable seizure detectors: A simulation study. *Epilepsia* 65, 4 (2024), 1017–1028. doi:10.1111/epi.17917

[31] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *Proceedings of the 34th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 70)*. 1321–1330. https://proceedings.mlr.press/v70/guo17a.html

[32] Nils Hammerla and Thomas Plötz. 2015. Let's (not) Stick Together: Pairwise Similarity Biases Cross-Validation in Activity Recognition. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. ACM, 1041–1051. doi:10.1145/2750858.2807551

[33] Kevin Anthony Hoff and Masooda Bashir. 2015. Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust. *Human Factors* 57, 3 (2015), 407–434. doi:10.1177/0018720814547570

[34] Giorgi Japaridze, Dirk Loeckx, Tim Buckinx, Sidsel Armand Larsen, Renée Proost, Katrien Jansen, Paul MacMullin, Natalia Paiva, Sofia Kasradze, Alexander Rotenberg, Lieven Lagae, and Sándor Beniczky. 2022. Automated detection of absence seizures using a wearable electroencephalographic device: a phase 3 validation study and feasibility of automated behavioral testing. *Epilepsia* (2022). doi:10.1111/epi.17200

[35] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*. https://proceedings.neurips.cc/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf

[36] Predrag Klasnja, Shawna Smith, Nicholas J. Seewald, Andy Lee, Kelly Hall, Brook Luers, Eric B. Hekler, and Susan A. Murphy. 2019. Efficacy of Contextually Tailored Suggestions for Physical Activity: A Micro-randomized Optimization Trial of HeartSteps. *Annals of Behavioral Medicine* 53, 6 (2019), 573–582. doi:10.1093/abm/kay067

[37] Utkarsh Kunwar, Sheetal Borar, Moritz Berghofer, Julia Kylmälä, Ilhan Aslan, Luis A. Leiva, and Antti Oulasvirta. 2022. Robust and Deployable Gesture Recognition for Smartwatches. In *Proceedings of the 27th International Conference on Intelligent User Interfaces (IUI '22)*. ACM, 277–291. doi:10.1145/3490099.3511125

[38] Florian Künzler, Varun Mishra, Jan-Niklas Kramer, David Kotz, Elgar Fleisch, and Tobias Kowatsch. 2019. Exploring the State-of-Receptivity for mHealth Interventions. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 3, 4, Article 140 (2019). doi:10.1145/3369805

[39] Gierad Laput and Chris Harrison. 2019. Sensing Fine-Grained Hand Activity with Smartwatches. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. ACM. doi:10.1145/3290605.3300568 Paper 338.

[40] John D. Lee and Katrina A. See. 2004. Trust in Automation: Designing for Appropriate Reliance. *Human Factors* 46, 1 (2004), 50–80. doi:10.1518/hfes.46.1.50_30392

[41] Ying Lei, Yancheng Cao, Will Ke Wang, Yuanzhe Dong, Changchang Yin, Weidan Cao, Ping Zhang, Jingzhen Yang, Bingsheng Yao, Yifan Peng, Chunhua Weng, Randy Auerbach, Lena Mamykina, Dakuo Wang, Yuntao Wang, and Xuhai Xu. 2026. WatchGuardian: Enabling User-Defined Personalized Just-in-Time Intervention on Smartwatch. *ACM Transactions on Computing for Healthcare* (2026). doi:10.1145/3788689 Design details in this paper are taken from the preprint, arXiv:2502.05783.

[42] K. W. Leibinger, E. Murray, S. Aschenbrenner, and J. Randerath. 2023. Short-term intervention complemented by wearable technology improves Trichotillomania – A naturalistic single-case report. *Frontiers in Psychology* 14 (2023), 1071532. doi:10.3389/fpsyg.2023.1071532

[43] Hong Li, Shishir Chawla, Richard Li, Sumeet Jain, Gregory D. Abowd, Thad Starner, Cheng Zhang, and Thomas Plötz. 2018. WristWash: Towards Automatic Handwashing Assessment Using a Wrist-worn Device. In *Proceedings of the 2018 ACM International Symposium on Wearable Computers (ISWC '18)*. ACM, 132–139. doi:10.1145/3267242.3267247

[44] Aleksej Logacjov, Kerstin Bach, Atle Kongsvold, Hilde Bremseth Bårdstu, and Paul Jarle Mork. 2021. HARTH: A Human Activity Recognition Dataset for Machine Learning. *Sensors* 21, 23 (2021), 7853. doi:10.3390/s21237853

[45] Natalia Martinez Gil, Fearghal O'Donncha, Wesley Gifford, Nianjun Zhou, Dhaval Patel, and Roman Vaculin. 2026. Adaptive Conformal Anomaly Detection with Time Series Foundation Models for Signal Monitoring. In *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. https://research.ibm.com/publications/adaptive-conformal-anomaly-detection-with-time-series-foundation-models-for-signal-monitoring

[46] Taylor R. Mauldin, Marc E. Canby, Vangelis Metsis, Anne H. H. Ngu, and Coralys Cubero Rivera. 2018. SmartFall: A Smartwatch-Based Fall Detection System Using Deep Learning. *Sensors* 18, 10 (2018), 3363. doi:10.3390/s18103363

[47] Sinziana Mazilu, Ulf Blanke, Michael Hardegger, Gerhard Tröster, Eran Gazit, Moran Dorfman, and Jeffrey M. Hausdorff. 2014. GaitAssist: A Wearable Assistant for Gait Training and Rehabilitation in Parkinson's Disease. In *2014 IEEE International Conference on Pervasive Computing and Communication Workshops (PerCom Workshops)*. IEEE, 135–137. doi:10.1109/PerComW.2014.6815179

[48] Sinziana Mazilu, Ulf Blanke, and Gerhard Tröster. 2015. Gait, Wrist, and Sensors: Detecting Freezing of Gait in Parkinson's Disease from Wrist Movement. In *2015 IEEE International Conference on Pervasive Computing and Communication Workshops (PerCom Workshops), Workshop on Sensing Systems and Applications Using Wrist Worn Smart Devices*. IEEE,

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

579–584.

[49] A. Mehrotra, V. Pejovic, J. Vermeulen, R. J. Hendley, and M. Musolesi. 2016. My Phone and Me: Understanding People's Receptivity to Mobile Notifications. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. ACM, 1021–1032. doi:10.1145/2858036.2858566

[50] P. Meritam, P. Ryvlin, and S. Beniczky. 2018. User-based evaluation of applicability and usability of a wearable accelerometer device for detecting bilateral tonic-clonic seizures: A field study. *Epilepsia* 59, S1 (2018), 48–52. https://serval.unil.ch/notice/serval:BIB_3420901C7730

[51] Annamaria Mesaros, Toni Heittola, and Tuomas Virtanen. 2016. Metrics for Polyphonic Sound Event Detection. *Applied Sciences* 6, 6 (2016), 162. doi:10.3390/app6060162

[52] Joachim Meyer. 2004. Conceptual Issues in the Study of Dynamic Hazard Warnings. *Human Factors* 46, 2 (2004), 196–204. https://cris.bgu.ac.il/en/publications/conceptual-issues-in-the-study-of-dynamic-hazard-warnings/

[53] Vimal Mollyn, Karan Ahuja, Dhruv Verma, Chris Harrison, and Mayank Goel. 2022. SAMoSA: Sensing Activities with Motion and Subsampled Audio. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 6, 3, Article 132 (2022). doi:10.1145/3550284

[54] Md Abu Sayeed Mondol and John A. Stankovic. 2015. Harmony: A Hand Wash Monitoring and Reminder System Using Smart Watches. In *Proceedings of the 12th EAI International Conference on Mobile and Ubiquitous Systems (MobiQuitous)*. 11–20.

[55] Inbal Nahum-Shani, Shawna N. Smith, Bonnie J. Spring, Linda M. Collins, Katie Witkiewitz, Ambuj Tewari, and Susan A. Murphy. 2018. Just-in-Time Adaptive Interventions (JITAIs) in Mobile Health: Key Components and Design Principles for Ongoing Health Behavior Support. *Annals of Behavioral Medicine* 52, 6 (2018), 446–462. doi:10.1007/s12160-016-9830-8

[56] Francesco Onorati, Giulia Regalia, Chiara Caborni, Matteo Migliorini, Daniel Bender, Cherise Frazier, Eliana Kovitch Thropp, Elizabeth D. Mynatt, Jonathan Bidwell, Roberto Mai, W. Curt LaFrance, Andrew S. Blum, Daniel Friedman, Tobias Loddenkemper, Fatemeh Mohammadpour-Touserkani, Claus Reinsberger, Simone Tognetti, Ming Zher Poh, Rosalind W. Picard, et al. 2017. Multicenter clinical assessment of improved wearable multimodal convulsive seizure detectors. *Epilepsia* (2017). https://dspace.mit.edu/handle/1721.1/111959

[57] Luca Palmerini, Jochen Klenk, Clemens Becker, and Lorenzo Chiari. 2020. Accelerometer-Based Fall Detection Using Machine Learning: Training and Testing on Real-World Falls. *Sensors* 20, 22 (2020), 6479. doi:10.3390/s20226479

[58] Abhinav Parate, Meng-Chieh Chiu, Chaniel Chadowitz, Deepak Ganesan, and Evangelos Kalogerakis. 2014. RisQ: Recognizing Smoking Gestures with Inertial Sensors on a Wristband. In *Proceedings of the 12th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys '14)*. ACM, 149–161. doi:10.1145/2594368.2594379

[59] V. Pejovic and M. Musolesi. 2014. InterruptMe: Designing Intelligent Prompting Mechanisms for Pervasive Applications. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '14)*. ACM. doi:10.1145/2632048.2632062

[60] Martin Pielot and Luz Rello. 2017. Productive, Anxious, Lonely: 24 Hours Without Push Notifications. In *Proceedings of the 19th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '17)*. ACM. doi:10.1145/3098279.3098526

[61] Parisa Pour Rezaei, Tero Jokela, Akos Vetek, and Marja Salmimaa. 2024. Informing the Design of Intervention Solutions for Body-Focused Repetitive Behaviors. *Proceedings of the ACM on Human-Computer Interaction* 8, MHCI (2024). doi:10.1145/3676513

[62] Samaneh Rezaeimanesh, Mohsen Behradfar, Mohammad Fili, and Guiping Hu. 2026. Deep Multimodal Wearable Sensor Fusion for Detection of Body-Focused Repetitive Behaviors. arXiv:2608.09830 https://arxiv.org/abs/2608.09830

[63] Stephen Rice. 2009. Examining Single- and Multiple-Process Theories of Trust in Automation. *Journal of General Psychology* 136, 3 (2009), 303–322. doi:10.3200/GENP.136.3.303-322

[64] Alina Roitberg, Monica Haurilet, Manuel Martinez, and Rainer Stiefelhagen. 2021. Uncertainty-sensitive Activity Recognition: a Reliability Benchmark and the CARING Models. arXiv:2101.00468 https://arxiv.org/abs/2101.00468

[65] Debaditya Roy, Sarunas Girdzijauskas, and Serghei Socolovschi. 2021. Confidence-Calibrated Human Activity Recognition. *Sensors* 21, 19 (2021), 6566. doi:10.3390/s21196566

[66] Mrinmoy Roy, Venkata Devesh Reddy Seethi, Rami Lake, and Pratool Bharti. 2022. CovidAlert – A Wristwatch-based System to Alert Users from Face Touching. In *Pervasive Computing Technologies for Healthcare (EAI, Springer LNICST)*. Springer. doi:10.1007/978-3-030-99194-4_30

[67] Nazir Saleheen, Amin Ahsan Ali, Syed Monowar Hossain, Hillol Sarker, Soujanya Chatterjee, Benjamin Marlin, Emre Ertin, Mustafa al'Absi, and Santosh Kumar. 2015. puffMarker: A Multi-Sensor Approach for Pinpointing the Timing of First Lapse in Smoking Cessation. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. ACM, 999–1010. https://digitalcommons.memphis.edu/facpubs/19836

[68] Benjamin Lucas Searle, Dimitris Spathis, Marios Constantinides, Daniele Quercia, and Cecilia Mascolo. 2021. Anticipatory Detection of Compulsive Body-focused Repetitive Behaviors with Wearables. In *Proceedings of the 23rd*

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

International Conference on Mobile Human-Computer Interaction (MobileHCI '21). ACM. doi:10.1145/3447526.3472061

[69] Jan Slemenšek, Jelka Geršak, Božidar Bratina, Vesna Marija van Midden, Zvezdan Pirtošek, and Riko Šafarič. 2024. Wearable Online Freezing of Gait Detection and Cueing System. *Bioengineering* 11, 10 (2024), 1048. doi:10.3390/bioengineering11101048

[70] Andrés Tello, Victoria Degeler, and Alexander Lazovik. 2024. Too Good To Be True: performance overestimation in (re)current practices for Human Activity Recognition. In *2024 IEEE International Conference on Pervasive Computing and Communications Workshops (PerCom Workshops)*. IEEE. doi:10.1109/PerComWorkshops59983.2024.10503465

[71] Edison Thomaz, Irfan Essa, and Gregory D. Abowd. 2015. A Practical Approach for Recognizing Eating Moments with Wrist-Mounted Inertial Sensing. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. ACM. doi:10.1145/2750858.2807545

[72] Xin Tong, Yang Feng, and Jingyi Jessica Li. 2018. Neyman-Pearson Classification Algorithms and NP Receiver Operating Characteristics. *Science Advances* 4, 2 (2018), eaao1659. doi:10.1126/sciadv.aao1659

[73] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. 2022. *Algorithmic Learning in a Random World* (2 ed.). Springer Cham. doi:10.1007/978-3-031-06649-8

[74] Fei Wang, Tingting Zhang, Xilei Wu, Pengcheng Wang, Xin Wang, Han Ding, Jingang Shi, Jinsong Han, and Dong Huang. 2025. You Can Wash Hands Better: Accurate Daily Handwashing Assessment with a Smartwatch. *IEEE Transactions on Mobile Computing* (2025). doi:10.1109/TMC.2025.3571805

[75] Jamie A. Ward, Paul Lukowicz, and Hans W. Gellersen. 2011. Performance Metrics for Activity Recognition. *ACM Transactions on Intelligent Systems and Technology* 2, 1, Article 6 (2011). doi:10.1145/1889681.1889687

[76] Gary M. Weiss, Kenichi Yoneda, and Thaier Hayajneh. 2019. Smartphone and Smartwatch-Based Biometrics Using Activities of Daily Living. *IEEE Access* 7 (2019), 133190–133202. doi:10.1109/ACCESS.2019.2940729

[77] Christopher D. Wickens and Stephen R. Dixon. 2007. The Benefits of Imperfect Diagnostic Automation: A Synthesis of the Literature. *Theoretical Issues in Ergonomics Science* 8, 3 (2007), 201–212. doi:10.1080/14639220500370105

[78] Taeyoung Yeon, Vasco Xu, Henry Hoffmann, and Karan Ahuja. 2025. WatchHAR: Real-time On-device Human Activity Recognition System for Smartwatches. In *Proceedings of the 27th International Conference on Multimodal Interaction (ICMI '25)*. ACM. doi:10.1145/3716553.3750775

[79] Shuai Zhang, Chuan Zhou, Yang Liu, Peng Zhang, Xixun Lin, and Shirui Pan. 2025. Conformal Anomaly Detection in Event Sequences. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. 77177–77193. https://mlanthology.org/icml/2025/zhang2025icml-conformal/

A Proof of Proposition 4.1 and Remarks

Setting. Let $\mathcal{G} = \{g_1 < \cdots < g_G\}$ be a finite grid whose last point g_G never fires, so $\mathrm{FNH}(g_G) = 0$. The detector, the trigger policy (k, R) and $\mathcal{G}$ are fixed before the calibration and test users are seen. For user j let $V_j = \tau_j^\star$ be the certified threshold defined in Section 4.1. Index the calibration users $1, \ldots, n$ and the new user $n+1$. Users are exchangeable given the training users, and V_j is the same deterministic function of each user's data, so $(V_1, \ldots, V_{n+1})$ is exchangeable.

Claim. For $m \le n$, let $\hat{\tau} = V_{(m)}$ be the m-th smallest of $V_1, \ldots, V_n$. Then $\Pr(V_{n+1} > \hat{\tau}) \le 1 - m/(n+1)$.

Proof. If $V_{n+1} > V_{(m)}$, then at least m of the other n values lie strictly below V_{n+1}. Write $N_j = \#\{l \le n+1 : V_l < V_j\}$. The event implies $N_{n+1} \ge m$. Sort the $n+1$ values. Only values in sorted positions $m+1, \ldots, n+1$ can have at least m values strictly below them, so $\sum_{j=1}^{n+1} \mathbf{1}\{N_j \ge m\} \le n+1-m$ for every realisation, ties included. By exchangeability, $\Pr(N_{n+1} \ge m) = \frac{1}{n+1}\mathbb{E}\sum_j \mathbf{1}\{N_j \ge m\} \le \frac{n+1-m}{n+1}$. Hence $\Pr(V_{n+1} \le \hat{\tau}) \ge m/(n+1) \ge 1-\delta$ for $m = \lceil (n+1)(1-\delta) \rceil$.

Finally, $\hat{\tau} \in \mathcal{G}$. If $V_{n+1} \le \hat{\tau}$, the definition of V_{n+1} as the start of the monotone envelope gives $\mathrm{FNH}_{n+1}(\tau') \le B$ for every grid threshold $\tau' \ge V_{n+1}$, in particular for $\tau' = \hat{\tau}$. □

Remarks.

(1) **Ties.** The finite grid makes ties among the V_j common. The counting argument above already covers ties, which only make the bound conservative. Without ties, $\Pr(V_{n+1} > \hat{\tau}) = 1 - m/(n+1)$ exactly. The probability of exceeding the budget can still be smaller, because the envelope is conservative: $V_{n+1} > \hat{\tau}$ does not imply $\mathrm{FNH}_{n+1}(\hat{\tau}) > B$.

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

(2) **Grid and discretisation.** The guarantee concerns grid thresholds. Deploying a threshold that is not in $\mathcal{G}$, or moving it online, is not covered: the envelope says nothing between grid points.
(3) **Never firing.** If $m > n$, or if $\hat{\tau} = g_G$, BCT returns a threshold that never fires. The bound then holds trivially, with zero recall. We treat this as "cannot certify at a useful operating point" and report it, rather than counting it as a success.
(4) **The floor, the conditional violation rate and sessions.** The floor is discussed in Section 4.1, the violation rate conditional on one calibration set in Section 4.2, and what a session and a new wearer mean in Section 3.3.

B Datasets and Integrity Checks

CAPTURE-24 came from its official archive (Oxford University Research Archive, https://doi.org/10.5287/bodleian:NGx0JOMP5, CC BY 4.0). WISDM-2019, HARTH and Daphnet came from public GitHub mirrors, because the original hosts were unreachable from our compute environment, and we checked each mirror against the published facts below.

- **CAPTURE-24** [16]. We downloaded the release (a 6.9 GB archive) and processed all 151 participants: 3,883 h of recording (the data paper reports 3,883 h). Annotations follow the release's label dictionary. Windows whose majority annotation maps to one of our 11 classes cover 2,349 h (the data paper reports 2,562 h annotated, before our 5 s windowing and the exclusion of ambiguous eating codes).
- **WISDM-2019** [76]. We counted 3,777,046 watch-accelerometer and 3,440,342 watch-gyroscope rows. Together with the phone files, the total is exactly the 15,630,426 rows of the UCI release. We removed the duplicated rows of one subject. Eleven subject–activity pairs are missing from the original release, four subjects were recorded at about 50/25 Hz, and some pairs contain 100 Hz bursts; higher-rate recordings were anti-aliased before resampling to 20 Hz.
- **Daphnet** [6]. We count 237 freeze episodes and 499.4 minutes of recording, both matching the original paper. Label 0 marks time outside the experiment and was removed. Two subjects have no freezes. Sensor dropouts were left in the data, since they occur in deployment too.
- **HARTH** [44]. The mirror has 31 subjects and 46.2 h of recording. This is HARTH v2.0. Its first 22 files contain exactly the 6,461,328 rows of UCI v1.2, which our citation describes; the 9 later subjects could be checked only against the repository. Three undocumented label codes (about 3 minutes in total) were kept as non-target time.

C CAPTURE-24 Results by Task and Budget

Table 5 gives the values behind Figure 5, with median onset latencies and the recall of the in-sample per-user oracle.

Submitted manuscript (anonymous review version), under review at PACM IMWUT. Not yet peer-reviewed.

Table 5. CAPTURE-24 by task and budget ($\delta = 0.1$; 10 splits, 51 test participants each): share of new participants over budget with two-way cluster-bootstrap 95% intervals, mean event recall, and median onset latency, for the fixed threshold and BCT, and the recall of the in-sample per-user oracle. The finite-sample bound for BCT is 9.8% in every row.

		Fixed $\tau = 0.5$			BCT ($\delta = 0.1$)			Oracle
Task	B/h	over [95% CI]	recall	lat.	over [95% CI]	recall	lat.	recall
Eating	1	31.4% [22.3, 43.8]	0.45	101 s	8.6% [4.5, 14.8]	0.27	143 s	0.41
	2	12.7% [7.1, 20.4]	0.45	101 s	6.9% [3.1, 12.5]	0.39	125 s	0.44
	4	3.7% [0.7, 8.1]	0.45	101 s	3.7% [0.7, 8.1]	0.45	101 s	0.45
In vehicle	1	25.1% [15.6, 34.5]	0.93	66 s	9.2% [4.5, 14.8]	0.91	76 s	0.93
	2	10.4% [4.9, 17.0]	0.93	66 s	8.0% [3.6, 13.7]	0.93	69 s	0.94
	4	3.3% [0.6, 7.4]	0.93	66 s	2.7% [0.4, 6.6]	0.93	68 s	0.94
Bicycling	1	2.2% [0.4, 4.3]	0.88	57 s	2.2% [0.4, 4.3]	0.88	57 s	0.95
	2	0.2% [0.0, 1.1]	0.88	57 s	0.2% [0.0, 1.1]	0.88	57 s	0.95
	4	0.0% [0.0, 0.0]	0.88	57 s	0.0% [0.0, 0.0]	0.88	57 s	0.95
Sports	1	1.6% [0.2, 3.5]	0.16	95 s	1.6% [0.2, 3.5]	0.16	95 s	0.17
	2	0.8% [0.0, 2.6]	0.16	95 s	0.8% [0.0, 2.6]	0.16	95 s	0.17
	4	0.2% [0.0, 1.1]	0.16	95 s	0.2% [0.0, 1.1]	0.16	95 s	0.17